GRACE NAKIYEMBA

SOFTWARE ENGINEER | PYTHON & DJANGO | AI & DIGITAL SYSTEMS | OPEN-SOURCE | BLOCKCHAIN & WEB3

Kampala, Uganda | +256 756 339 110 | gracenakiyemba256@gmail.com

LinkedIn | GitHub | Portfolio

PROFESSIONAL PROFILE

Uganda-based software engineer developing practical web applications and digital workflows using Python, Django, React, REST APIs, and relational databases. Experience spans business operations platforms, low-connectivity USSD/SMS services, network monitoring, collaborative open-source work, and technical project coordination. Combines engineering delivery with training, public communication, data administration, and leadership; interested in impactful software and AI-enabled products.

TECHNICAL & PROFESSIONAL SKILLS

Programming & Web: Python, JavaScript, SQL, HTML, CSS, C++; Django, Django REST Framework, FastAPI, React, Vite.

Backend & Delivery: REST API design and integration, authentication, role-based access control, PostgreSQL, SQLite, Git/GitHub, Docker, automated tests, debugging, technical documentation.

Applied Technologies: Africa's Talking SMS/USSD integrations, AI-assisted workflows and responsible AI, IoT/embedded systems learning, blockchain/Web3 studies.

Operations & Collaboration: Microsoft Office, Google Workspace, research, data entry, records management, Agile/Scrum, stakeholder communication, facilitation, leadership.

PROFESSIONAL EXPERIENCE

Software Engineer | Independent & Collaborative Projects | 2025–Present

- Build database-backed business applications and API integrations; translate real-world requirements into maintainable features, permissions, and workflows.
- Develop and test backend components, investigate defects, document changes, and collaborate through Git and GitHub.

Python Development Intern | Codveda Technologies (Remote) | Mar–Jun 2025

- Completed Python projects including a blog, student management system, task manager, and calculator; practiced data handling, testing, debugging, and independent delivery.

Data Entry & Examination Support | Uganda National Examinations Board | Two years; shelving in 2025

- Supported examination data processing with emphasis on accuracy, confidentiality, deadlines, and established procedures; assisted with examination-material shelving in 2025.

Data Entry & Sales Records | SB Veterinary Drugshop, Rubindi, Mbarara | 2020–2021

- Recorded sales and other business information and maintained organized operational records.

Office Assistant | Faculty of Civil Engineering, Kyambogo University | 2021–2022

- Assisted with document organization, filing, routine data entry, and office administration.

Technical Content Creator & Community Communicator | LinkedIn | Jan 2025–Present

- Create accessible technology and career-development content and engage a professional audience of more than 10,000 followers.

Social Media Manager | CTK Worship Team | 2024–2025

- Planned content, prepared digital communications, and supported promotion of activities and events.

SELECTED ENGINEERING & HACKATHON PROJECTS

Project Nyondo | Hardware Stock & Sales Management | Django, SQL

- Developed role-based stock, sales, supplier-credit, and deposit workflows for Admin, Store Manager, and Sales Attendant; deployed for use by a small hardware-business team.

NetSage | Network Monitoring & Incident Response | Django, PostgreSQL, React

- Contributed APIs and data models for sites, devices, telemetry, alerts, incidents, and notifications; worked on database constraints and automated tests.

Tuviora | Accessible Event Operations Platform | Django, Africa's Talking | In development

- Building event-management and notification workflows, including SMS integration and USSD-oriented access for low-connectivity participants.

AmaniCare | Women's Health Information via USSD/SMS | Africa's Talking Hackathon, 2026

- Co-developed an accessible service for feature-phone users; project recognized with first place in the Women in Tech hackathon.

Sauti Yo | Multi-channel Legal Information Platform | Africa's Talking Hackathon, 2026

- Contributed Django backend content models and seed data for rights information across USSD, SMS, voice, chatbot, and web; project placed second.

PlayConnect | Multi-channel Community Platform | Africa's Talking Hackathon, 2026

- Collaborated on a React/FastAPI/Gemini solution integrating web and telecom channels; team placed second.

KaCyber SafeMove | Cybersecurity-focused Hackathon Project | Code Queen Cohort 14, 2026

- Served as Scrum Master for an eight-person team, coordinating delivery of a first-place hackathon project.

OPEN-SOURCE CONTRIBUTIONS

OpenDataEnsemble (ODE) / Community Days | Contributor | 2026

- Investigated issues, developed focused code changes and regression coverage, and collaborated through pull requests and maintainer review.
- Worked on responsive desktop layout behavior and app-bundle test fixtures; submitted pull requests for maintainer review.

LEADERSHIP, TEACHING & COMMUNITY IMPACT

- Class President, Groundbreaker Talents Cohort 6 (2026–Present): Represent and coordinate communication for approximately 40 learners.
- STEAM / AI Learning Facilitator, Women Ascension / WALUAIS (2026): Prepared and delivered sessions on systems thinking, responsible AI, data and analytics, and project development.
- Build Her Tech Mentorship Programme (2026): Co-facilitated mentorship and peer-learning sessions supporting women entering technology.
- Public speaking: International Women's Day 2026 (GDG Cloud Mbarara / Women Techmakers Mbarara); International Day of Women and Girls in Science 2026 (Women Ascension UK).

EDUCATION & PROFESSIONAL DEVELOPMENT

- Software & AI Engineering | Turing College | Jul 2026–Expected Jan 2027 (in progress)
- Certificate in Software Engineering with Python | Refactory Academy | Feb–Jun 2026
- UACE | St. Jude SSS Kyabakuza | 2019
- UCE | Baptist High School | 2017

CERTIFICATIONS & ADDITIONAL LEARNING

- Aspire Leaders Program | Aspire Institute | 2026
- Project Planning & Management | University of Virginia | 2026

- Web Design & Development with JavaScript; AI for Entrepreneurship | Code Queen
- Personal Branding | Coursera | 2026
- Additional learning: blockchain technologies (BSV, Hedera, Ethereum); IoT and embedded systems, including C++ and sensor-based demonstrations.

SELECTED RECOGNITION

First-place recognition: AmaniCare (Africa's Talking Women in Tech) and KaCyber SafeMove (Code Queen Cohort 14). Second-place recognition: Sauti Yo and PlayConnect (Africa's Talking). Featured by Ellevate Amsterdam in connection with Groundbreaker Talents participation.

REFEREES

Wanyana Liz Prossy | Programs Lead, Educating The Children | lizprossy@milerconsult.com | +256 780 248 377

Nansubuga Esther | Programs Manager, Groundbreaker Talents | esther@groundbreaker.org | +256 703 679 786